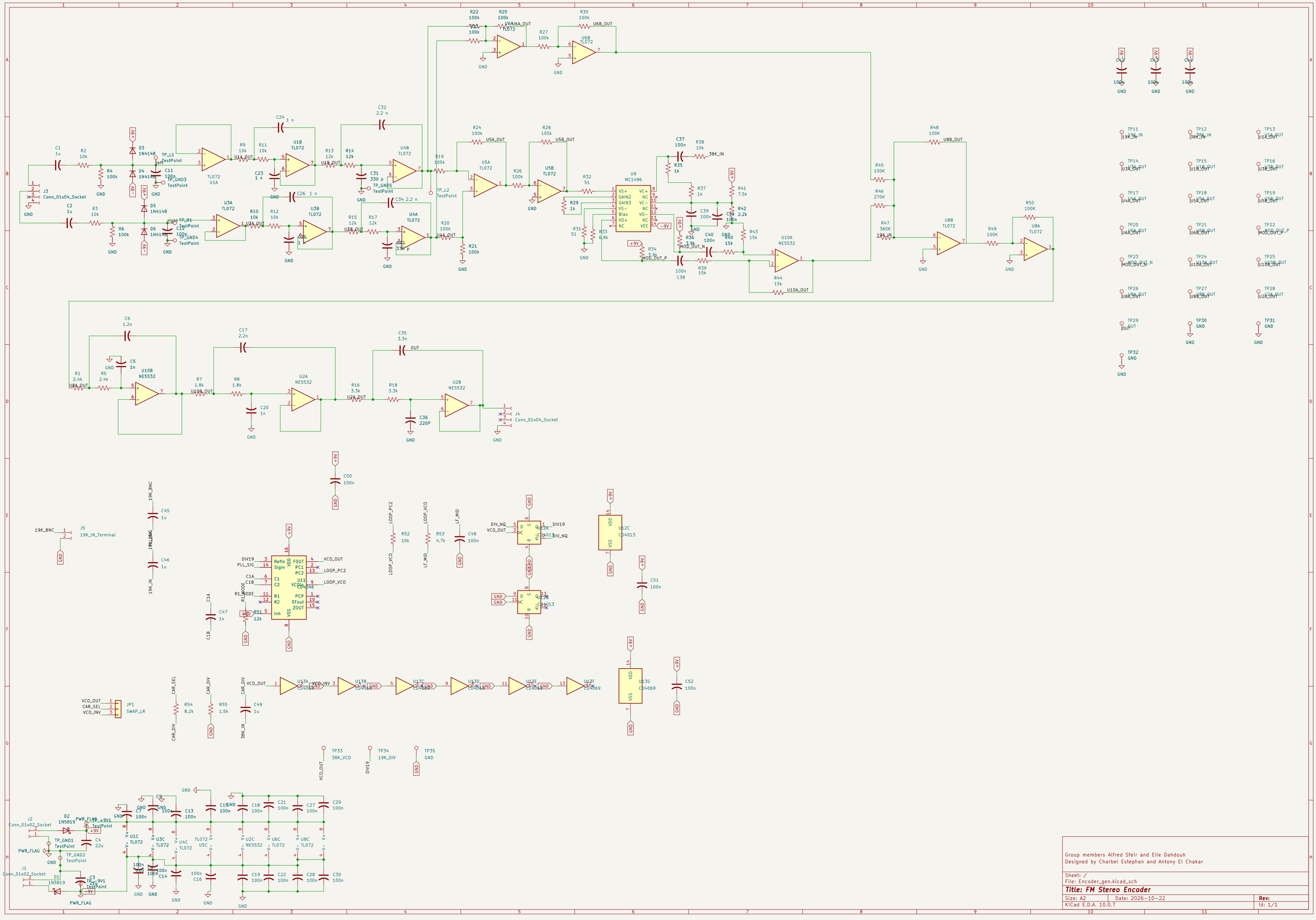

J3
Conn_01x04_Socket
C1 1u
C2 1u
R1 2.4k
R2 10k
R3 10k
R4 100k
R5 2.4k
R6 100k
D3 1N4148
D4 1N4148
D5 1N4148
D6 1N4148
C11 100p
C12 100p
TP_L1
TP_R1
TP_GND3
TP_GND4
U1A TL072
U3A TL072
R9 10k
R10 10k
U1A_OUT
U3A_OUT
C23 1 n
C24 1 n
C26 1 n
R11 10k
R12 10k
U1B TL072
U3B TL072
R13 12k
R14 12k
R15 12k
R17 12k
U1B_OUT
U3B_OUT
C31 330 p
C32 2.2 n
C34 2.2 n
C21 330 p
U4A TL072
U4B TL072
TP_GND5
TP_GND6
TP_L2
R19 100k
R20 100k
U4A_OUT
R21 100k
R22 100k
R23 100k
U6A TL072
U6A_OUT
R25 100k
R27 100k
R30 100k
U6B TL072
U6B_OUT
R24 100k
U5A_OUT
U5A TL072
R26 100k
R28 100k
U5B_OUT
U5B TL072
R29 1k
R32 51
R31 51
R33 6.8k
U9 MC1496
VS+
VS-
Bias
VO+
NC
GAIN2
GAIN3
VC+
VC-
VO-
VEE
+9V
-9V
C37 100n
R38 10k
38K_IN
R35 1k
R37 1k
C39 100n
C41 100n
R41 7.5k
R42 2.2k
R36 3.9k
MOD_OUT_N
R34 3.9k
MOD_OUT_P
C40 100n
C38 100n
R40 15k
R39 15k
R43 15k
R44 15k
U10A NE5532
U10A_OUT
R45 100K
R48 100K
U8B_OUT
R46 270K
R47 360K
19K_IN
U8B TL072
R49 100K
R50 100K
U8A TL072
C6 1.2n
C5 1n
U10B NE5532
U8A_OUT
R7 1.8k
U10B_OUT
R8 1.8k
C20 1n
C17 2.2n
U2A NE5532
R16 3.3k
U2A_OUT
C35 3.3n
R18 3.3k
C36 220P
U2B NE5532
J4
Conn_01x04_Socket
J5
19K_IN_Terminal
19K_BNC
C45 1u
C46 1u
C47 1n
C50 100n
R52 10k
R53 4.7k
C48 100n
LOOP_PC2
LOOP_VCO
LF_MID
R51 12k
R_MODE
C1A
C1B
U11 CD4046
Refin
SigIn
VCOin
DemOut
R1
R2
PCP
SFout
ZOUT
Inh
VDD
VSS
FOUT
PC1
PC2
U12A CD4013
U12B CD4013
U12C CD4013
C51 100n
C52 100n
U13A CD4069
U13B CD4069
U13C CD4069
U13D CD4069
U13E CD4069
U13F CD4069
U13G CD4069
VCO_OUT
VCO_INV
CAR_SEL
CAR_DIV
JP1
SWAP_LR
R54 8.2k
R55 1.5k
C49 1u
TP33 38K_VCO
TP34 19K_DIV
TP35
J2
J1
D2 1N5819
D1 1N5819
PWR_FLAG
C4 22u
C3 22u
C7 100n
C8 100n
C13 100n
C15 100n
C18 100n
C21 100n
C27 100n
C29 100n
C14 100n
C16 100n
C19 100n
C22 100n
C28 100n
C30 100n
U1C TL072
U3C TL072
U4C TL072
U5C TL072
U2C NE5532
U6C TL072
U8C TL072
TP_GND1
TP_GND2
TP_9V1
TP_9V2
TP11 19K_IN
TP12 38K_IN
TP13 U1A_OUT
TP14 U3A_OUT
TP15 U1B_OUT
TP16 U3B_OUT
TP17 U4A_OUT
TP18 U4B_OUT
TP19 U5B_OUT
TP20 U6A_OUT
TP21 U6B_OUT
TP22 MOD_OUT_P
TP23 MOD_OUT_N
TP24 U10A_OUT
TP25 U10B_OUT
TP26 U8A_OUT
TP27 U8B_OUT
TP28 U2A_OUT
TP29 OUT
TP30
TP31
TP32
Group members Alfred Sfeir and Elie Dahdouh
Designed by Charbel Estephan and Antony El Chakar
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File: Encoder_gen.kicad_sch
Title: FM Stereo Encoder
Size: A2
Date: 2026-10-22
KiCad E.D.A. 10.0.7
Rev:
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